System Identification

What it is

- Process of figuring out the mathematical model of our system, or robot, so that we can design a PID controller
- Usually use the WPIlib System Identification Tool to build a mathematical model
- It gives us a model that we can predict the outcome when given a situation such as how fast the motor will go when 5 volts is applied.
- The mathematical model is not “perfect” as there are many measurement noise(timing error) and system noise(unmodeled force of the system like vibrations). But it is “good enough” to give us an accurate feedforward control
- To determine numeric values for each coefficient in our model, a curve-fitting technique (such as least-squares regression) is applied to measurements taken from the real mechanism.

Identification Tools

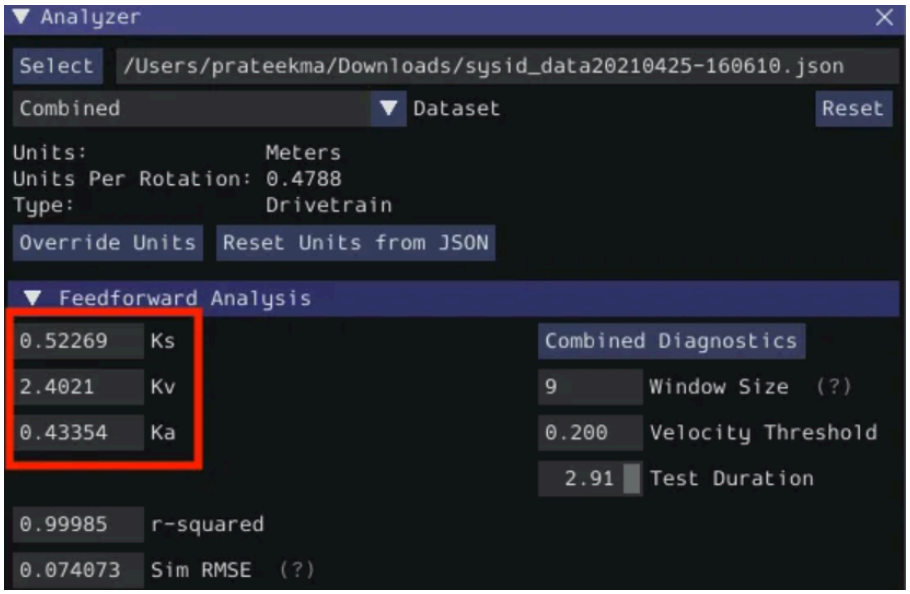
- Simple Motor Identification
- $$V = K_s \cdot sgn(\dot{d}) + K_v \cdot \dot{d} + K_a \cdot \ddot{d}$$
- where V is the applied voltage, d(w a dot) is its velocity, and d(w two dots) is its acceleration.
- the model for a permanent-magnet dc motor with no loading other than friction and inertia
- **The most basic Model for SysID**

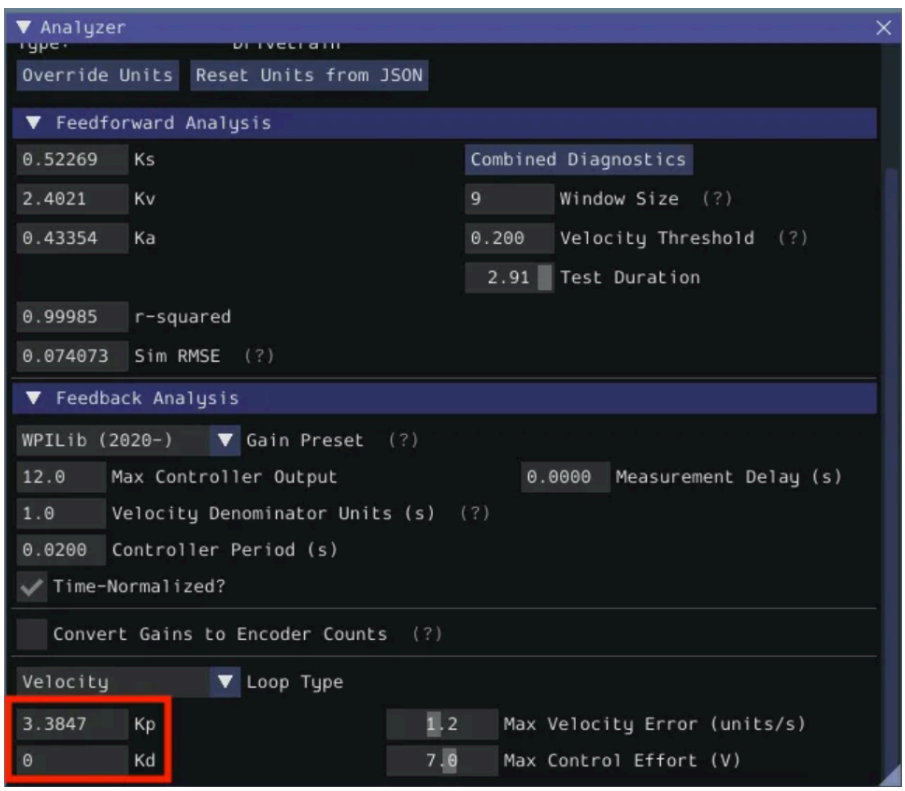
- Elevator Identification
- $$V = K_g + K_s \cdot sgn(\dot{d}) + K_v \cdot \dot{d} + K_a \cdot \ddot{d}$$
- Where V is the applied voltage, d is the displacement of the elevator, d(w a dot) is its velocity, and d(w two dots)
- The constant Kg is added to correctly account for gravity

- Arm Identification
- $$V = K_g \cdot cos(\theta) + K_s \cdot sgn(\dot{\theta}) + K_v \cdot \dot{\theta} + K_a \cdot \ddot{\theta}$$
- Where V is the applied voltage, delta is the angular displacement of the arm, delta w a dot is its angular velocity, delta w two dots is its angular acceleration
- The cosine term is added to correctly account for gravity

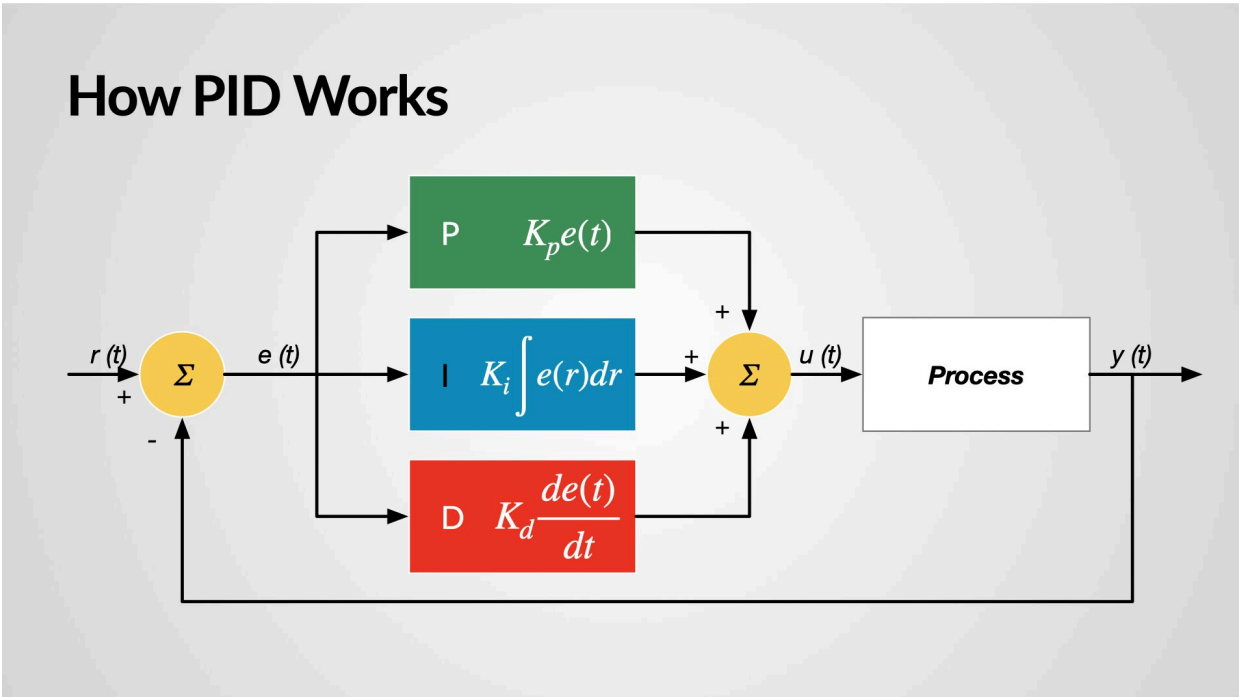
Types of Tests

- Quasistatic

    - the mechanism is gradually sped-up such that the voltage corresponding to acceleration is negligible
- Dynamic
    - a constant ‘step voltage’ is given to the mechanism, so that the behavior while accelerating can be determined.
- When we collect the data from our robot, we can get the Ks, Kv and Ka from the analysis and also the PID gains
- 

- 

PID Tuning



- Basically fixing errors of the output(feedback control)
- $$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt} e(t)$$
- Proportional
  - Reacts with current error, multiply the constant by the current error
  - High Kp - Faster, Stronger response, but can overshoot
- Integral
  - Reacts to the sum of the past errors, removes steady state errors
  - Too high Ki - can make oscillations and instability
- Derivative
  - Reacts with the Rate of change of error, Makes a prediction and prevents overshoot
  - High Kd - can make the system noisy and jittery

- Anti Windup Control
  - Needed when the error is so big due to saturation of the actuator and that the integral becomes too big
  - Keeps the integrated value from increasing past some specific limit

    - Clamping
        - Basically turning Integrator off
        - Have two routes
            - First comparing the input and output of the PID controller before and after the Saturation check, if not equal, there is a saturation so it stops the integration
            - Compare the sign of the output and input of the PID controller, if they are the same, they are being overshoot, so it stops the integration
        - If both routes outputs a 1, the integration stops
- Noise Filtering
    - Low Pass filter
        - Removes the fast and noisy spikes of the error